

Your grade: 100%

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Next item →

1. To compute $\hat{x}_k^-$, $\Sigma_{\hat{x},k}^-$, and $\hat{z}_k$, what information do we require? Select all that apply.

1 / 1 point

☒ We need to know model matrices A_{k-1} , B_{k-1} , C_k , and D_k .

☒ **Correct**
Yes. These matrices are needed in the prediction steps.

☒ We need to know the process-noise covariance matrix $\Sigma_{\tilde{w}}$.

☒ **Correct**
Yes. We need to know this information for step 1b.

☐ We need to know the sensor-noise covariance matrix $\Sigma_{\tilde{v}}$.

☒ We need to have an estimate of the prior values of the state vector, $\hat{x}_{k-1}^+$.

☒ **Correct**
Yes. This is needed in step 1a.

☒ We need to know the error covariance matrix of the prior state estimate, $\Sigma_{\hat{x},k-1}^+$.

☒ **Correct**
Yes. This is needed in step 1b.

2. To be able to compute $\hat{x}_k^+$ and $\Sigma_{\hat{x},k}^+$, what do we need to know? Select all that apply.

1 / 1 point

☒ The state prediction error covariance $\Sigma_{\hat{x},k}^-$.

☒ **Correct**
Yes. This is needed by steps 2a and 2c.

☒ The sensor-noise covariance matrix $\Sigma_{\tilde{v}}$.

☒ **Correct**
Yes. This is needed by step 2a.

☐ The process-noise covariance matrix $\Sigma_{\tilde{w}}$.

☒ The model matrices C_k and D_k .

☒ **Correct**
Yes. These are needed by steps 2a and 2c.

☒ The predicted value of the state at the present time, $\hat{x}_k^-$.

☒ **Correct**
yes. This is needed by step 2b.

3. What is the significance of the state estimation error covariance matrix $\Sigma_{\hat{x},k}^+$?

1 / 1 point

☐ It indicates the uncertainty of the predicted state.

☒ It allows computing error bounds on the state estimate produced by the Kalman filter.

☐ It is needed to compute the output prediction error $\tilde{z}_k = z_k - \hat{z}_k$.

☐ It is needed to be able to compute the Kalman gain matrix L_k .

☒ **Correct**
Yes. This is the primary significance and use of this matrix.